

Functions of Two or Three Real Variables (2022–2026)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS

26

TOTAL MARKS

44 M

TEST DURATION

78 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2022–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	11	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	7	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	8	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q52 • ID: JAM_2026_Q52)

NAT

+2 M

Neg: Nil (0)



Q.52

Let the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} 4x^2 \tan^{-1}\left(\frac{y}{2x}\right) + y^3 \tan^{-1}\left(\frac{x}{4y^2}\right), & xy \neq 0 \\ 0, & \text{otherwise} \end{cases}$$

Then $\left(\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x}\right)\right)$ at $(0,0)$ is _____ (rounded off to two decimal places).

Question 2 (JAM 2026 • Q46 • ID: JAM_2026_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46

Let $z = \cos(4x + 5y)$, where $x = \frac{\pi}{2} + 2\theta$, $y = -\left(\frac{\pi}{4} + \theta\right)$.Then $\left.\frac{dz}{d\theta}\right|_{\theta=\frac{\pi}{4}} = \text{_____}$ (rounded off to one decimal place).

Question 3 (JAM 2026 • Q36 • ID: JAM_2026_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36

$$\text{Let } f(x, y) = \begin{cases} \frac{x^2 y}{1+x^2} \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Which of the following statements is/are TRUE?

- (A) $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y)$ exists.
- (B) $\frac{\partial f}{\partial x}$ is continuous at the point $(0, 1)$.
- (C) $\frac{\partial f}{\partial y}$ is continuous at the point $(0, 1)$.
- (D) $\lim_{(x, y) \rightarrow (0, 0)} \frac{f(x, y)}{\sqrt{x^2 + y^2}}$ does not exist.

Question 4 (JAM 2026 • Q17 • ID: JAM_2026_Q17)

MCQ

+2 M

Neg: -0.67

Q.17	Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function defined by $f(x, y) = \begin{cases} \frac{5x^2y - 4y^3}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$ Which ONE of the following statements is FALSE?
(A)	f is continuous at $(0, 0)$
(B)	$\frac{\partial f}{\partial x}$ at $(0, 1) = 0$
(C)	$\frac{\partial f}{\partial y}$ at $(1, 0) = 5$
(D)	$\frac{\partial f}{\partial y}$ is continuous at $(0, 0)$

Question 5 (JAM 2026 • Q15 • ID: JAM_2026_Q15)

MCQ

+2 M

Neg: -0.67

Q.15	Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by $f(x, y) = (x^2 - 1)^2 + (y^2 - 1)^2 \text{ for all } (x, y) \in \mathbb{R}^2.$ Which ONE of the following statements is TRUE?
(A)	f has local maxima at exactly two points.
(B)	f has local minima at exactly two points.
(C)	f has local minima at exactly three points.
(D)	f has exactly four saddle points.

Question 6 (JAM 2025 • Q50 • ID: JAM_2025_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{(x^2 - y^2)xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Then, the value of $\frac{\partial f}{\partial y}(1, 0) - \frac{\partial f}{\partial x}(0, 2)$ is equal to _____ (rounded off to two decimal places).

Question 7 (JAM 2025 • Q34 • ID: JAM_2025_Q34)

MSQ

+2 M

Neg: Nil (0)

Q.34 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{(x^2 + \sin x)y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Then, which of the following is/are TRUE?

- (A) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists and $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 1$
- (B) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists and $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$
- (C) f is differentiable at $(0, 0)$
- (D) f is NOT differentiable at $(0, 0)$

Question 8 (JAM 2025 • Q31 • ID: JAM_2025_Q31)

MSQ

+2 M

Neg: Nil (0)

Q.31 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{xy^2 + y^5}{x^2 + y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Then, which of the following is/are TRUE?

- (A) The iterated limits $\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} f(x, y) \right)$ and $\lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} f(x, y) \right)$ exist
- (B) Exactly one of the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exists at $(0, 0)$
- (C) Both the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist at $(0, 0)$
- (D) f is NOT differentiable at $(0, 0)$

Question 9 (JAM 2025 • Q27 • ID: JAM_2025_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = e^y(x^2 + y^2) \quad \text{for all } (x, y) \in \mathbb{R}^2.$$

Then, which one of the following is TRUE?

- (A) The number of points at which f has a local minimum is 2
- (B) The number of points at which f has a local maximum is 2
- (C) The number of points at which f has a local minimum is 1
- (D) The number of points at which f has a local maximum is 1

Question 10 (JAM 2024 • Q35 • ID: JAM_2024_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35

Let

$$S = \{(x, y) \in \mathbb{R}^2: x > 0, y > 0\},$$

and $f: S \rightarrow \mathbb{R}$ be given by

$$f(x, y) = 2x^2 + 3y^2 - \log x - \frac{1}{6} \log y.$$

Then, which of the following statements is/are TRUE?

- (A) There is a unique point in S at which $f(x, y)$ attains a local maximum
- (B) There is a unique point in S at which $f(x, y)$ attains a local minimum
- (C) For each point $(x_0, y_0) \in S$, the set $\{(x, y) \in S: f(x, y) = f(x_0, y_0)\}$ is bounded
- (D) For each point $(x_0, y_0) \in S$, the set $\{(x, y) \in S: f(x, y) = f(x_0, y_0)\}$ is unbounded

Question 11 (JAM 2024 • Q30 • ID: JAM_2024_Q30)

MCQ

+2 M

Neg: -0.67

Q.30 Define the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = 12xy e^{-(2x+3y-2)}.$$

If (a, b) is the point of local maximum of f , then $f(a, b)$ equals

- (A) 2
- (B) 6
- (C) 12
- (D) 0

Question 12 (JAM 2024 • Q2 • ID: JAM_2024_Q2)

MCQ

+1 M

Neg: -0.33

Q.2 For a twice continuously differentiable function $g: \mathbb{R} \rightarrow \mathbb{R}$, define

$$u_g(x, y) = \frac{1}{y} \int_{-y}^y g(x+t) dt \quad \text{for } (x, y) \in \mathbb{R}^2, \quad y > 0.$$

Which one of the following holds for all such g ?

(A) $\frac{\partial^2 u_g}{\partial x^2} = \frac{2}{y} \frac{\partial u_g}{\partial y} + \frac{\partial^2 u_g}{\partial y^2}$

(B) $\frac{\partial^2 u_g}{\partial x^2} = \frac{1}{y} \frac{\partial u_g}{\partial y} + \frac{\partial^2 u_g}{\partial y^2}$

(C) $\frac{\partial^2 u_g}{\partial x^2} = \frac{2}{y} \frac{\partial u_g}{\partial y} - \frac{\partial^2 u_g}{\partial y^2}$

(D) $\frac{\partial^2 u_g}{\partial x^2} = \frac{1}{y} \frac{\partial u_g}{\partial y} - \frac{\partial^2 u_g}{\partial y^2}$

Question 13 (JAM 2023 • Q51 • ID: JAM_2023_Q51)

NAT

+2 M

Neg: Nil (0)

Q. 51 Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be defined as $f(x, y, z) = x^3 + y^3 + z^3$, and let $L: \mathbb{R}^3 \rightarrow \mathbb{R}$ be the linear map satisfying

$$\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{f(1+x, 1+y, 1+z) - f(1,1,1) - L(x,y,z)}{\sqrt{x^2 + y^2 + z^2}} = 0.$$

Then $L(1, 2, 4)$ is equal to _____. (rounded off to two decimal places)

Question 14 (JAM 2023 • Q41 • ID: JAM_2023_Q41)

NAT

+1 M

Neg: Nil (0)

Q. 41 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function defined as follows:

$$f(x, y) = \begin{cases} (x^2 - 1)^2 \cos^2 \left(\frac{y^2}{(x^2 - 1)^2} \right) & \text{if } x \neq \pm 1 \\ 0 & \text{if } x = \pm 1. \end{cases}$$

The number of points of discontinuity of $f(x, y)$ is equal to _____.

Question 15 (JAM 2023 • Q37 • ID: JAM_2023_Q37)

MSQ

+2 M

Neg: Nil (0)

Q. 37 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as follows:

$$f(x, y) = \begin{cases} \frac{x^4 y^3}{x^6 + y^6} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Then

(A) $\lim_{t \rightarrow 0} \frac{f(t, t) - f(0, 0)}{t}$ exists and equals $\frac{7}{2}$

(B) $\left. \frac{\partial f}{\partial x} \right|_{(0,0)}$ exists and equals 0

(C) $\left. \frac{\partial f}{\partial y} \right|_{(0,0)}$ exists and equals 0

(D) $\lim_{t \rightarrow 0} \frac{f(t, 2t) - f(0, 0)}{t}$ exists and equals $\frac{1}{3}$

Question 16 (JAM 2023 • Q36 • ID: JAM_2023_Q36)

MSQ

+2 M

Neg: Nil (0)

Q. 36 A subset $S \subseteq \mathbb{R}^2$ is said to be *bounded* if there is an $M > 0$ such that $|x| \leq M$ and $|y| \leq M$ for all $(x, y) \in S$. Which of the following subsets of $\mathbb{R}^2$ is/are bounded?

- (A) $\{(x, y) \in \mathbb{R}^2 : e^{x^2} + y^2 \leq 4\}$
- (B) $\{(x, y) \in \mathbb{R}^2 : x^4 + y^2 \leq 4\}$
- (C) $\{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 4\}$
- (D) $\{(x, y) \in \mathbb{R}^2 : e^{x^3} + y^2 \leq 4\}$

Question 17 (JAM 2023 • Q17 • ID: JAM_2023_Q17)

MCQ

+2 M

Neg: -0.67

Q. 17 Let $f(x, y) = \ln(1 + x^2 + y^2)$ for $(x, y) \in \mathbb{R}^2$. Define

$$P = \frac{\partial^2 f}{\partial x^2} \Big|_{(0,0)} \quad Q = \frac{\partial^2 f}{\partial x \partial y} \Big|_{(0,0)}$$

$$R = \frac{\partial^2 f}{\partial y \partial x} \Big|_{(0,0)} \quad S = \frac{\partial^2 f}{\partial y^2} \Big|_{(0,0)}$$

Then

- (A) $PS - QR > 0$ and $P < 0$
- (B) $PS - QR > 0$ and $P > 0$
- (C) $PS - QR < 0$ and $P > 0$
- (D) $PS - QR < 0$ and $P < 0$

Question 18 (JAM 2023 • Q16 • ID: JAM_2023_Q16)

MCQ

+2 M

Neg: -0.67

Q. 16 Let $f(x, y) = e^{x^2+y^2}$ for $(x, y) \in \mathbb{R}^2$, and a_n be the determinant of the matrix

$$\begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

evaluated at the point $(\cos(n), \sin(n))$. Then the limit $\lim_{n \rightarrow \infty} a_n$ is

- (A) non-existent
- (B) 0
- (C) $6e^2$
- (D) $12e^2$

Question 19 (JAM 2023 • Q8 • ID: JAM_2023_Q8)

MCQ

+1 M

Neg: -0.33

Q. 8 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $f(x, y) = (e^x \cos(y), e^x \sin(y))$. Then the number of points in $\mathbb{R}^2$ that do NOT lie in the range of f is

- (A) 0
- (B) 1
- (C) 2
- (D) infinite

Question 20 (JAM 2022 • Q56 • ID: JAM_2022_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56

Define $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = x^2 + 2y^2 - x \quad \text{for } (x, y) \in \mathbb{R}^2.$$

Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$ and $E = \{(x, y) \in \mathbb{R}^2 : \frac{x^2}{4} + \frac{y^2}{9} \leq 1\}$.

Consider the sets

$$D_{\max} = \{(a, b) \in D : f \text{ has absolute maximum on } D \text{ at } (a, b)\},$$

$$D_{\min} = \{(a, b) \in D : f \text{ has absolute minimum on } D \text{ at } (a, b)\},$$

$$E_{\max} = \{(c, d) \in E : f \text{ has absolute maximum on } E \text{ at } (c, d)\},$$

$$E_{\min} = \{(c, d) \in E : f \text{ has absolute minimum on } E \text{ at } (c, d)\}.$$

Then the total number of elements in the set

$$D_{\max} \cup D_{\min} \cup E_{\max} \cup E_{\min}$$

is equal to _____.

Question 21 (JAM 2022 • Q51 • ID: JAM_2022_Q51)

NAT

+2 M

Neg: Nil (0)

Q.51

For $t \in \mathbb{R}$, let $[t]$ denote the greatest integer less than or equal to t .

Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 4\}$. Let $f: D \rightarrow \mathbb{R}$ and $g: D \rightarrow \mathbb{R}$ be defined by $f(0, 0) = g(0, 0) = 0$ and

$$f(x, y) = [x^2 + y^2] \frac{x^2 y^2}{x^4 + y^4}, \quad g(x, y) = [y^2] \frac{xy}{x^2 + y^2}$$

for $(x, y) \neq (0, 0)$. Let E be the set of points of D at which both f and g are discontinuous. The number of elements in the set E is _____.

Question 22 (JAM 2022 • Q42 • ID: JAM_2022_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42

Consider the function $u: \mathbb{R}^3 \rightarrow \mathbb{R}$ given by

$$u(x_1, x_2, x_3) = x_1 x_2^4 x_3^2 - x_1^3 x_3^4 - 26x_1^2 x_2^2 x_3^3.$$

Let $c \in \mathbb{R}$ and $k \in \mathbb{N}$ be such that

$$x_1 \frac{\partial u}{\partial x_2} + 2x_2 \frac{\partial u}{\partial x_3}$$

evaluated at the point (t, t^2, t^3) , equals ct^k for every $t \in \mathbb{R}$. Then the value of k is equal to _____.

Question 23 (JAM 2022 • Q36 • ID: JAM_2022_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36

Let M be a positive real number and let $u, v: \mathbb{R}^2 \rightarrow \mathbb{R}$ be continuous functions satisfying

$$\sqrt{u(x, y)^2 + v(x, y)^2} \geq M\sqrt{x^2 + y^2} \quad \text{for all } (x, y) \in \mathbb{R}^2.$$

Let $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by

$$F(x, y) = (u(x, y), v(x, y)) \quad \text{for } (x, y) \in \mathbb{R}^2.$$

Then which of the following is/are true?

(A)

 F is injective.

(B)

If K is open in $\mathbb{R}^2$, then $F(K)$ is open in $\mathbb{R}^2$.

(C)

If K is closed in $\mathbb{R}^2$, then $F(K)$ is closed in $\mathbb{R}^2$.

(D)

If E is closed and bounded in $\mathbb{R}^2$, then $F^{-1}(E)$ is closed and bounded in $\mathbb{R}^2$.

Question 24 (JAM 2022 • Q15 • ID: JAM_2022_Q15)

MCQ

+2 M

Neg: -0.67

Q.15

For $t \in \mathbb{R}$, let $[t]$ denote the greatest integer less than or equal to t . Define functions $h: \mathbb{R}^2 \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ by

$$h(x, y) = \begin{cases} \frac{-1}{x^2 - y} & \text{if } x^2 \neq y, \\ 0 & \text{if } x^2 = y \end{cases} \quad \text{and} \quad g(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Then which one of the following is FALSE?

(A)

$$\lim_{(x, y) \rightarrow (\sqrt{2}, \pi)} \cos\left(\frac{x^2 y}{x^2 + 1}\right) = \frac{-1}{2}.$$

(B)

$$\lim_{(x, y) \rightarrow (\sqrt{2}, 2)} e^{h(x, y)} = 0.$$

(C)

$$\lim_{(x, y) \rightarrow (e, e)} \ln(x^{y - [y]}) = e - 2.$$

(D)

$$\lim_{(x, y) \rightarrow (0, 0)} e^{2y} g(x) = 1.$$

Question 25 (JAM 2022 • Q10 • ID: JAM_2022_Q10)

MCQ

+1 M

Neg: -0.33

Q.10

Let $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$. Consider the functions

$$u : \mathbb{R}^2 - \{(0,0)\} \rightarrow \mathbb{R} \quad \text{and} \quad v : \mathbb{R}^2 - \{(0,0)\} \rightarrow \mathbb{R}$$

given by

$$u(x,y) = x - \frac{x}{x^2 + y^2} \quad \text{and} \quad v(x,y) = y + \frac{y}{x^2 + y^2}.$$

The value of the determinant $\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ at the point $(\cos \theta, \sin \theta)$ is equal to

(A) $4 \sin \theta$.(B) $4 \cos \theta$.(C) $4 \sin^2 \theta$.(D) $4 \cos^2 \theta$.

Question 26 (JAM 2022 • Q9 • ID: JAM_2022_Q9)

MCQ

+1 M

Neg: -0.33

Q.9

Let c be a positive real number and let $u: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$u(x, t) = \frac{1}{2c} \int_{x-ct}^{x+ct} e^{s^2} ds \quad \text{for } (x, t) \in \mathbb{R}^2.$$

Then which one of the following is true?

(A)

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{on } \mathbb{R}^2.$$

(B)

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{on } \mathbb{R}^2.$$

(C)

$$\frac{\partial u}{\partial t} \frac{\partial u}{\partial x} = 0 \quad \text{on } \mathbb{R}^2.$$

(D)

$$\frac{\partial^2 u}{\partial t \partial x} = 0 \quad \text{on } \mathbb{R}^2.$$



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OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Functions of Two or Three Real Variables (2022–2026) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	0.24 to 0.26
2	NAT	+1	3.0 to 3.0
3	MSQ	+2	A;C
4	MCQ	+2	D
5	MCQ	+2	D
6	NAT	+1	2.98 to 3.02
7	MSQ	+2	B;D
8	MSQ	+2	A;C;D
9	MCQ	+2	C
10	MSQ	+2	B;C
11	MCQ	+2	A
12	MCQ	+1	A
13	NAT	+2	20.99 to 21.01

Q. No.	Type	Marks	Official Key
14	NAT	+1	0 to 0
15	MSQ	+2	A, B, C
16	MSQ	+2	A, B, C
17	MCQ	+2	B
18	MCQ	+2	D
19	MCQ	+1	B
20	NAT	+2	5 to 5
21	NAT	+2	18 to 18
22	NAT	+1	14 to 14
23	MSQ	+2	C,D
24	MCQ	+2	Marks to All
25	MCQ	+1	D
26	MCQ	+1	A

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.